Using Physics Informed Classical Machine Learning for the Detection of Exoplanets

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1 Introduction

2 Methodology

We use the Kepler Object of Interest (KOI) dataset to train our model.

2.1 Transit Depth Consistency

Since the telescope makes 2D measurements, we assert that the stellar flux must be directly related to the area of the star that is observed. During a transit, a portion of the area of the star is blocked by the planet, leading to a decrease in the observed flux. The transit depth, which is the fractional decrease in flux during a transit, can be expressed as:

$$\delta_{\text{theoretical}} = \frac{\text{Area}_{\text{Planet}}}{\text{Area}_{\text{Star}}} = \frac{\pi R_p^2}{\pi R_*^2} \tag{1}$$

where R_p is the radius of the planet and R_* is the radius of the star. We can compare this theoretical transit depth with the observed transit depth from the light curve data to ensure consistency.

$$\Delta\delta = \left|\delta_{obs} - \delta_{\text{theoretical}}\right| \tag{2}$$

2.2 Duration Consistency

2.3 Impact Parameter

3 Results

Table 1: 10-Fold Stratified Cross-Validation Results by Dataset and Feature Set. Note: KOI and TESS utilized complete cases, while K2 utilized KNN-imputed data.

Dataset	**Feature Set**	**Mean Accuracy**	**Mean PR-AUC**
KOI (k=10)	Ablation Features	0.9467 ± 0.0074	0.9905 ± 0.0019
	Features 3	0.9550 ± 0.0066	0.9913 ± 0.0027
	Features 4	0.9544 ± 0.0064	0.9918 ± 0.0023
TESS (k=10)	Ablation Features	0.8481 ± 0.0082	0.5729 ± 0.0405
	Features 3	0.8477 ± 0.0107	0.5774 ± 0.0491
	Features 4	0.8478 ± 0.0126	0.5933 ± 0.0451
K2 Imputed (k=10)	Ablation Features	0.7948 ± 0.0239	0.9022 ± 0.0149
	Features 3	0.7948 ± 0.0275	0.8957 ± 0.0118
	Features 4	0.7879 ± 0.0290	0.8855 ± 0.0148

Table 2: K2 Dataset Performance: Native NaN Handling vs. KNN Imputation (10-Fold CV)

Model & Data Strategy	**Feature Set**	**Mean Accuracy**	**Mean PR-AUC**
HistGB (Native NaN)	Ablation Features	0.8380 ± 0.0174	0.9375 ± 0.0096
	Features 3	0.8246 ± 0.0161	0.9297 ± 0.0104
	Features 4	0.8102 ± 0.0181	0.9046 ± 0.0184
RandomForest (Imputed)	Ablation Features	0.7963 ± 0.0241	0.9022 ± 0.0167
	Features 3	0.7971 ± 0.0291	0.8962 ± 0.0168
	Features 4	0.7946 ± 0.0277	0.8870 ± 0.0197

Table 3: TESS Dataset Performance: Native NaN Handling vs. KNN Imputation (10-Fold CV)

Model & Data Strategy	**Feature Set**	**Mean Accuracy**	**Mean PR-AUC**
HistGB (Native NaN)	Ablation Features	0.8575 ± 0.0126	0.5543 ± 0.0493
	Features 3	0.8602 ± 0.0108	0.5569 ± 0.0496
	Features 4	0.8609 ± 0.0083	0.5591 ± 0.0404
RandomForest (Imputed)	Ablation Features	0.8590 ± 0.0113	0.5445 ± 0.0471
	Features 3	0.8602 ± 0.0091	0.5431 ± 0.0457
	Features 4	0.8610 ± 0.0076	0.5608 ± 0.0430